level

Ap-1 : Log + Rev + Compare.

$TC = O(n) + O(n) + O(n) = O(n)$.

$AS = O(n)$

Ap-2 2-ptr

$TC = O(n)$

$AS = O(1)$

race a car $\longrightarrow$ race a car

A man; nana $\longrightarrow$ amannanam ✓

Lowercase	Uppercase	Digit
ch >= 'a' &&	ch >= 'A' &&	ch >= '0' &&
ch <= 'z'	ch <= 'Z'	ch <= '9'

Uppercase to lowercase

ch - 'A' + 'a'

Lowercase to Uppercase

ch - 'a' + 'A'

1) Same. $\boxed{ch1 == ch2}$

2) ch1 is Uppercase ch2 is lowercase (if the same char). $\boxed{ch1 - 'A' + 'a' == ch2}$

3) ch1 is lowercase ch2 is uc. (if the same char) $\boxed{ch1 == ch2 - 'A' + 'a'}$

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101 ✓

1010 ✗

27332 ✗

22322 ✓

$n = 1283$

$res = 8321$

$n/10 = 128$!

$res = res * 10 + (n/10)$

$n = n/10$

$T(n) = n + (2n-8) + (4n-16) + (8n-32) + \cdots$

$T(n) = n + 2(n-1) + 4(n-2) + 8(n-3) + 16(n-4) + \cdots$

$2T(n) = 0 + 2n + 4(n-1) + 8(n-2) + 16(n-3) + \cdots$

$2T(n) - T(n) = -n + 2 + 4 + 8 + 16 + \cdots$

$T(n) = -n + 2\left(\frac{2^n - 1}{2 - 1}\right)$

$= -n + 2^{n+1} - 2$

$= O(2^{n+1}) = O(2^n)$

Power (a, b)

$a = 2$

$b = 3$.

$res = 8$

$a = 3$

$b = 2$

$res = 9$

Ap-1

$res = 1$

for (i=0; i< b; i++)

$res = res * a$

$TC = O(b)$

$AS = O(1)$

Ap-1 $F(a, b) = a * F(a, b-1) \Rightarrow T(b) = T(b-1) + c$

$TC = O(b)$

$AS = O(b)$

$\boxed{8}$

$\left.\begin{matrix} a=2, & b=3 \end{matrix}\right\} 8$

$\left.\begin{matrix} a=2, & b=3 \end{matrix}\right\} 4$

$\left.\begin{matrix} a=2, & b=2 \end{matrix}\right\} 2$

$\left.\begin{matrix} a=2, & b=1 \end{matrix}\right\} 1$

$\boxed{a = 2, \ b = 0}$

Ap-3 $F(a, b) = \left[F\left(a, \frac{b}{2}\right)\right]^2$ if b is even.

$a * \left[F\left(a, \frac{b}{2}\right)\right]^2$ if b is odd.

$T(b) = T\left(\frac{b}{2}\right) + c$] $TC = O(\log b)$, $AS = O(\log b)$

$T(b)$
$\downarrow$
$T(b/2)$
$\downarrow$
$T(b/4)$
$\downarrow$
$T(b/8)$
$\vdots$
$T(0)$

$\Big\}$ $H = \log b$

$b = 8$

$\frac{8}{2} = 4$

$a^8 = a^4 \times a^4$

$b = 10$

$\frac{10}{2} = 5$.

$a^{10} = a^5 \times a^5$

Count digits in a Num

$n = 10123$

$TC = O(\log n)$

$AS = O(1)$

Count digits in all nums from 1 to n

Ap-1 $TC = O(n \log_{10} n)$

$AS = O(1)$

1
2
3
4
5
6
7
8
9
10
11
12
13
14
15
16
17
18
19
100

Ap-2

No of digits in 1st place $= n - (1-1)$

No of digits in 10th place $= n - 9 = n - (10-1)$

" " " " 100th place $= n - 99 = n - (100-1)$

100th $= n - 99 + n - (1000-1)$

$i = 1$; ____ ; $i^* = 10$

$n = 25$.

$res = res + [n - (i-1)]$

$i \to$ till 100

$25 - (1-1) = 25$

$25 - (10-1) = 16$

$[25 - (100-1)] = 25 - 99$ ✗

$n - (i-1) > 0$

$n > (i-1)$

$\boxed{i \le n+1}$